

Mark schemes

Q1.

(a) $P(1) = \frac{1}{3}$

Allow AWRT 0.333

B1

1

(b) (i) $P(\text{both give 1}) = \frac{1}{9}$

Allow AWRT 0.111; ft c's answer to (a)

M1A1F

2

(ii) $P(\text{neither gives 1}) = \frac{4}{9}$

*M1 for correct use of $P(A') = 1 - P(A)$;
Allow AWRT 0.444*

M1A1

2

[5]

Q2.

(a) (i) $0.15 \times 0.30 = 0.045$
0.045 cao

B1

1

(ii) $0.25 \times (0.18 + 0.24) = 0.105$

method for Hughes 2 or more

M1

0.25 times their Hughes 2 or more

M1

0.105 cao

A1

3

(iii) $0.15 \times 0.24 + 0.25 \times (0.18 + 0.24)$
 $+ 0.20 \times (0.28 + 0.18 + 0.24) + 0.4 = 0.681$

reasonable attempt at enumerating possibilities

			M1	
		<i>correct expression for at least 2 possibilities</i>		
			M1	
		<i>completely correct method – allow 1 slip</i>		
			m1	
		<i>A1 0.681 cao</i>		
			A1	
				4
(b)	(i)	$0.4 + 0.15 \times 0.24 = 0.436$		
		<i>reasonable attempt to enumerate possibilities</i>		
			M1	
		<i>completely correct method</i>		
			m1	
		<i>0.436 cao</i>		
			A1	
				3
	(ii)	$0.15 \times (0.28 + 0.18 + 0.24)$ $+ 0.25 \times (0.18 + 0.24) = 0.21$		
		<i>reasonable attempt to enumerate possibilities</i>		
			M1	
		<i>correct expression for one (out of 2) possibilities</i>		
			M1	
		<i>completely correct method</i>		
			m1	
		<i>0.21 cao</i>		
			A1	
				4

[15]

Q3.

- (a) Binomial $n = 6$ $p = 0.3$
Binomial

B1

$P(2 \text{ or fewer}) = 0.744$
 $n = 6$ $p = 0.3$

B1

0.744 (0.744 to 0.745)

B1

- (b) $P(> 3) = 1 - P(3 \text{ or fewer})$
 $P(> 3) = 1 - P(3 \text{ or fewer}) \text{ or equivalent}$

M1

$$= 1 - 0.925$$

$$= 0.0705$$

0.0705 (0.07 to 0.071)

A1

- (c) $P(6) = 1 - 0.9993 = 0.0007$

M1

0.0007 (0.0007 to 0.0008)

A1

7

[7]

Q4.

- (a) 0.3

0.3 *cao*

B1

1

- (b) (i) $\frac{10}{30} = \frac{1}{3}$
 $\frac{1}{3}$ *acf*

B1

1

- (ii) $\frac{1}{3} \times 0.06 = 0.02$
Method - their (b) (i)

M1

1

- (iii) $\frac{15}{30} (0.30 + 0.15) = 0.225$
Method - generous

	0.225	M1	
		A1	2
(iv)	$\frac{1}{3} \times 0.06 + \frac{1}{2} \times 0.15 + \frac{5}{30} \times 0.18 = 0.125$ <i>Attempt at P(4* comedy) + P(4* drama) + P(4* other)</i>	M1	
	<i>Completely correct method</i>		
	0.125 cao	m1	
		A1	3
(v)	$\frac{3}{4} \times (0.20 + 0.35) + \frac{1}{4} \times (0.40 + 0.10) = 0.5375$ <i>Reasonable attempt</i>	M1	
	<i>Completely correct method</i>		
	0.5375 (0.537 to 0.538)	m1	
		A1	3
(c)	$3 \times \frac{10}{30} \times \frac{9}{29} \times \frac{15}{28} = 0.166$ 3	B1	
	<i>Allow omission of or incorrect "3" – allow with replacement</i>		
	0.166 (0.166 to 0.1665)	M1	
		A1	3

[14]

Q5.

(a)
$$\frac{1}{4} \times \frac{1}{3} \times \frac{1}{5} = \frac{1}{60} \text{ or } 0.167 \text{ or } 16.7\%$$

			M1	
	<i>For product</i>			
			A1	2
(b)	$1 - \frac{1}{60} = \frac{59}{60}$ or 0.983 or 98.3%		B1	1
(c)	$\left[\frac{1}{4} \times \frac{2}{3} \times \frac{4}{5} \right] + \left[\frac{3}{4} \times \frac{1}{3} \times \frac{4}{5} \right] + \left[\frac{3}{4} \times \frac{2}{3} \times \frac{1}{5} \right]$ <i>For 3 products attempted</i>			
	<i>For sum</i>		M1	
			M1	
	$= \frac{8}{60} + \frac{12}{60} + \frac{6}{60} = \frac{26}{60} \text{ or } \frac{13}{30}$ or 0.433 or 43.3%		A1	3
(d)	$\frac{6}{26} \text{ or } \frac{3}{13}$ or 0.231 or 23.1% <i>M1 for use of 26 as denominator</i>		M1 A1	2
[8]				

Q6.

(a)	0.3			
	<i>0.3 cao</i>			
			B1	1
(b)	$0.6 \times 0.3 \times 0.75 = 0.135$ <i>3 probabilities multiplied</i> <i>Correct method</i> <i>0.135 cao</i>			
			M1 m1 A1	3



- (c) (i) $0.4 \times 0.3 + 0.12$
method – may be earned in (ii)
0.12 cao

M1 A1

- (ii) $0.75 \times 0.4 = 0.3$
0.3 cao

A1

3

- (d) $0.5 \times 0.12 + 0.5 \times 0.3 = 0.21$
Use of 0.5
Correct method
0.21 cao

M1 m1 A1

3

[10]

Q7.

- (a) $0.65 \times 0.46 = 0.299$

M1 A1

2

- (b) $0.65 + 0.52 - 0.30 = 0.87$
For $0.65 + 0.52$

M1 A1

2

- (c) $1 - (0.52 + 0.38 - 0.25) = 0.35$
For $0.52 + 0.38 - 0.25$
For sensible effort at 1–

M1M1A1

3

- (d) $\frac{0.65 \times 0.46}{0.38} = 0.787$
For numerator
ft part (a), (not 0.65×0.38) M1 A1 3
For denominator

M1

[10]

Q8.



(a) (i) $0.2 \times 0.2 = 0.04$

Method

M1

0.04 cao acf

A1

(ii) $2 \times 0.2 \times 0.8 = 0.32$

2

B1

0.2 × 0.8 used

M1

0.32 cao acf

A1

5

(b) (i) $0.65^3 = 0.275$

0.275 (0.274 to 0.275) acf

B1

(ii) $3 \times 0.2^2 \times 0.8 = 0.096$

3

B1

0.2² × 0.8

B1

M1

0.096 cao acf

A1

(iii) $6 \times 0.15 \times 0.65 \times 0.2 = 0.117$

6

B1

Correct method (allow 3 instead of 6)

M1

0.117 cao acg

A1

7

[12]

Q9.



(a) (i) $P(S) = \frac{57}{174} = \frac{19}{58} = 0.328$

B1

1

(ii) $P(T) = \frac{18}{174} = \frac{3}{29} = 0.103$

B1

1

(iii) $P(R \cup T) = \frac{64}{174} = \frac{32}{87} = 0.368$
for attempt at 28 + 13 + 18 + 1 + 4

M1 A1

2

(iv) $P(H \cap R') = \frac{16}{174} = \frac{8}{87} = 0.0920$

for attempt at 1 + 4 + 4 + 7

M1

A1

2

(v) $P(S | T) = \frac{1}{18} = 0.0556$
for 1

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M1

for 18

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M1

A1

3

(iv) $P(R | H') = \frac{28}{127} = 0.220$
for 28

M1

for 127

M1

sc 0.221

A1

3

(b) $P(R \cap H \cap T) = \frac{13}{174} = 0.0747$
 for 13

M1 A1

Male aged 55 – 74 years (with high blood pressure)
 who is undergoing treatment
Not “given”

B1

3

[15]

Q10.

(a) $0.6^3 = 0.216$

M1

0.216 cao acf

A1

2

(b) $1 - 0.6^3 - 0.4^3$

Attempt at $1 - P(GGG) - P(LLL)$ oe

M1

Completely correct method

m1

$= 0.72$

0.72 cao acf

A1

3

(c)
$$\begin{array}{l} L | LGL \quad 0.4 \times 0.6 \times 0.4 \\ L | GLL \quad + \quad 0.6 \times 0.4 \times 0.4 \end{array}$$

Reasonable attempt at enumerating possibilities

M1

Correct method

m1

$= 0.192$

0.192 cao

A1

3

(d) 0.4

M1A1

2

- (e)
- | | | |
|----------|--------------------------|----|
| L GGG | 0.6 ³ | |
| L LGGG | + 0.4 × 0.6 ³ | |
| L GLGG | + 0.4 × 0.6 ³ | m1 |
| L GGLG | + 0.4 × 0.6 ³ | |

Reasonable attempt at enumerating possibilities

M1

Reasonable attempt at evaluating probability

m1

Completely correct method

m1

= 0.475

0.475 (0.475 to 0.4755)

A1

4

[14]

Q11.

- (a) (i) $P(B \cup C) = P(B) + P(C) - P(B \cap C)$
 $= P(B) + P(C) - P(B) \times P(C)$

For attempt at formula

M1

For independent events

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M1

sc M2 if $P(C) = 0.28 - 0.1 + 0.02$

Let $x = P(C)$

$0.28 = 0.1 + x - (0.1x)$

For attempt at $x = P(C)$

M1

so $0.18 = 0.9x$ $x = 0.2$

*For showing $P(C) = 0.2$ or using 0.2
and demonstrating 0.28*

ag

A1

4



$$(ii) \quad P(C' \cup B') = P(C') + P(B') - P(C' \cap B') \\ = 0.8 + 0.9 - (0.8 \times 0.9)$$

M1

For attempt at formula

A1

2

$$(b) \quad (i) \quad 0.1 \times 0.7 \times 0.8 = 0.056$$

M1 A1

2

$$(ii) \quad (i) \quad + (0.9 \times 0.7 \times 0.2)$$

M1

$$+ (0.9 \times 0.3 \times 0.8)$$

M1

$$= 0.056 + 0.126 + 0.216 = 0.398$$

A1

3

$$(iii) \quad \frac{0.216}{0.398} = 0.543$$

For 0.216 f.t.

M1f.t.

For 0.398 f.t.

M1f.t.

A1

3

[14]

Q12.

$$(a) \quad (i) \quad 0.8 \times 0.7 = 0.56$$

0.56 cao

B1

$$(ii) \quad 0.2 \times 0.3 = 0.06$$

Method

M1

0.06 cao

A1



(iii) $0.8 \times 0.3 + 0.2 \times 0.7 = 0.38$

Method - allow small slip

M1

(or $1 - 0.56 - 0.06 = 0.38$)

0.38 cao

A1

5

(b) (i) $0.8 \times 0.7 \times 0.95 = 0.532$

0.532 cao

B1

(ii) $0.8 \times 0.3 \times 0.95 + 0.2 \times 0.7 \times 0.15 \dots$

Attempt at $P(2) + P(3)$ or equivalent

M1

$\dots + 0.8 \times 0.7 \times 0.05 + 0.532 = 0.809$

Reasonable attempt at evaluating $P(2)$ (or $P(1)$ if relevant)

M1

Completely correct method

m1

0.809 cao

A1

5

[10]

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Q13.

(a) (i) Binomial $n = 20$ $p = 0.15$

Binomial $n = 20$ $p = 0.15$

B1

$P(0) = 0.0388$

0.0388 (0.0387 – 0.039)

B1

(ii) $P(1 \text{ or more}) = 1 - 0.0388 = 0.961$

M1

$P(1 \text{ or more}) = 1 - P(0)$

A1

4

$$0.961 \text{ (} 0.961 - 0.962 \text{)}$$

(b) (i) Binomial

B1

$$n = 50 \quad p = 0.08$$

$$B(50, 0.08)$$

B1

$$P(2 \text{ or more}) = 1 - 0.0827$$

P (2 or more) = 1 – P (1 or fewer) - may be earned in (ii)

M1

$$= 0.9173$$

(ii) $P(3 \text{ or more}) = 1 - 0.2260 = 0.774$

0.917 (0.917 – 0.918) and
0.774 (0.7735 – 0.7745)

A1

4

(c) $P(1 \text{ or more exec}) \times P(2 \text{ or more stand})$

Reasonable attempt to define ways of seats being available

M1

Correct ways of seats being available

m1

+ $P(0 \text{ exec}) \times P(3 \text{ or more stand})$

Method for probability, depends on previous M mark only in (c)

M1

Completely correct method (dependent on Ms in (a) and (b) and (c))

m1

$$0.9612 \times 0.9173 + 0.0388 \times 0.774 = 0.912$$

0.912 (0.91 - 0.914)

A1

5

[13]

Q14.

(a) (i) $0.4 \times 0.7 = 0.28$

Method

M1

2

0.28 cao

A1

(ii) $(1 - 0.4)(1 - 0.35) = 0.39$

Method (answer given)

M1

0.39 cao

A1

2

(iii) $1 - 0.28 - 0.39 = 0.33$

Method

M1

0.33 cao

A1

2

(b) (i) $0.4 \times 0.7 \times 0.55 = 0.154$

Method

M1

0.154 cao

A1

2

(ii) $0.6 \times 0.35 \times 0.25 = 0.0525$

Method

M1

0.0525 (0.052 – 0.053)

A1

2

[10]

Q15.

(a) $\frac{3}{22} = 0.136$

$\frac{3}{22}$ or $0.136 - 0.137$

B1

1

(b) $\frac{10}{22} \times \frac{9}{21} \times \frac{8}{20} = \frac{6}{77} = 0.0779$

Method - allow with replacement

M1

Completely correct method

m1

$\frac{8}{77}$ acf or $(0.0779 - 0.078)$

A1

3

(c) $\frac{10}{22} \times \frac{9}{21} \times \frac{8}{20} + \frac{6}{22} \times \frac{5}{21} \times \frac{4}{20} + \frac{6}{22} \times \frac{5}{21} \times \frac{4}{20}$

Add 3 probabilities

M1

Correct method - allow with replacement

m1

Correct method

m1

$= \frac{8}{77} = 0.104$

$\frac{8}{77}$ acf or $(0.103 - 0.104)$

A1

4

(d) $3 \times \frac{10}{22} \times \frac{9}{21} \times \frac{12}{20} = \frac{27}{77}$ or 0.351

B1

Correct method - allow with replacement/omit or incorrect 3

M1

Correct method

m1

$\frac{27}{77}$ acf $(0.35 - 0.351)$

A1
4

(e) $6 \times \frac{1}{22} \times \frac{9}{21} \times \frac{12}{20} = \frac{27}{385} = 0.0701$

B1
6

*Correct method - allow with replacement
/ omit or incorrect 6.*

M1

Correct method

m1

$\frac{27}{385}$ or $(0.07 - 0.0702)$

A1
4

[16]

Q16.

(a) (i) $\frac{1}{3}$ (0.33 or better)
Method

M1

(ii) 0.4
Method

M1

(iii) 0.4
*Clearly incorrect method M0
A1 all answers acf*

M1 A1
4

(b) (i) W, X
W, X

B1
1

(ii) V, W
V, W

B1

Not mutually exclusive – same member may be female and adult
Reason not mutually exclusive – can be obtained for V, X

E1

Not independent $P(V|W) = \frac{5}{24} \neq P(V)(\frac{1}{3})$
 $P(V|W) \neq P(V)$

E1

3

[8]

Q17.

- (a) $0.2 \times 0.3 \times 0.1 = 0.006$ (or 0.6%)
Product of “not going” intended

M1 A1

2

- (b) $0.8 \times 0.3 \times 0.1 = 0.024$ (or 2.4%)
Correct product intended

M1 A1

2

- (c) $0.024 + (0.2 \times 0.7 \times 0.1) + (0.2 \times 0.3 \times 0.9)$
Attempt at other “one only goes” probs

M1

$$= 0.024 + 0.014 + 0.054$$

Summation of three

m1

$$= 0.092 \text{ (or 9.2\%)}$$

A1

3

- (d) $\frac{0.024}{0.092} = 0.261$ (or $\frac{6}{23}$ or 26.1%)

Numerator

Denominator

(M1, M1, A0 for 0.26)

M1 M1 A1

3

[10]

Q18.

(a) (i) $0.25 \times 0.25 = 0.0625$
Method
 $0.0625 (0.062 - 0.063)$

M1/A1

(ii) $2 \times 0.25 \times 0.75 = 0.375$
 2
 0.25×0.75
 0.375 cao acf

M1/M1/A1

5

(b) (i) $\frac{8}{12} \times \frac{7}{11} = 0.424 \left(\frac{14}{33} \right)$
Method – allow with replacement
 $0.424(0.424 - 0.425)acf$

M1/A1

2

(ii) $3 \times \frac{8}{12} \times \frac{7}{11} \times \frac{4}{10} = 0.509 \left(\frac{28}{55} \right)$
 3
 $\frac{8}{12} \times \frac{7}{11} \times \frac{4}{10}$
 $0.509 (0.509 - 0.51)acf$

M1/M1/A1

3

(iii) $1 - \frac{8}{12} \times \frac{7}{11} \times \frac{6}{10} \times \frac{5}{9} = 0.859 \left(\frac{85}{99} \right)$
 $1 - P(0) \text{ or equivalent}$
 correct method
 $0.859(0.858 - 0.86)acf$

M1/m1/A1

3

[13]

Q19.

(a) $\frac{1}{31}$ or 0.0323
 $0.0322 - 0.0323 \text{ any correct form}$

B1

1

(b) $\left(\frac{1}{31}\right)^2 = \frac{1}{961}$ or 0.00104
0.001 – 0.00105 any correct form

M1 A1

2

(c) $\left(\frac{1}{31}\right)^2 = \frac{1}{961}$ or 0.00104
0.001 – 0.00105 any correct form

M1 A1

2

(d) $\frac{30}{31} \times \frac{29}{31} = \frac{870}{961}$

*method - allow $\left(\frac{30}{31}\right)^2$
 completely correct method*

M1 M1

= 0.905

0.904 – 0.906 any correct form

A1

3

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[8]

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Q20.

(a) $0.8 \times 0.3 \times 0.4 = 0.096$

M1 A1

2

(b) $0.8 \times 0.7 \times 0.6 = 0.336$

M1 A1

2

(c) $0.096 + 0.336 + (0.2 \times 0.7 \times 0.4) = 0.488$
*Amy hitting
 addition*

M1 M1 A1

3



(d) $P(B \mid \text{exactly one}) = \frac{0.096}{0.488}$
numerator
denominator
ft 'their' (a) and (c) if sensible

M1 M1 A1

3

[10]

Q21.

(a) (i) $0.6 \times 0.9 \times 0.9 \times 0.9 \times 0.45$
method for on time - any day
correct method - allow method for late every day

B1 M1

$= 0.197$
 $0.196 \sim 0.197$

A1

(ii) $1 - 0.197 = 0.803$
ft their answer to (i)

A1ft

4

(b) (i) 0.6×0.95

M1

$= 0.57$

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A1

(ii) $0.4 \times 0.3 + 0.6 \times 0.05$
method for one way
correct method

M1 M1

$= 0.15$

cao

A1

5

[9]

Q22.

(a) $0.18 \times 0.8 = 0.144$ or $\frac{18}{125}$

M1 A1

2

(b) $(0.144) + (0.82 \times 0.1) = 0.226$
for 0.82×0.1
for addition

M1 M1

or $\frac{113}{500}$

A1

3

(c) $\frac{(0.82 \times 0.1)}{0.226} = 0.363$ or $\frac{41}{113}$
for numerator
for denominator can
ft if sensible

M1 M1 A1

3

[8]

Q23.

(a) (i) $\frac{1}{8} = 0.125$
cao acf

B1

(ii) $\frac{7}{12} = 0.583$
0.583 (0.583 – 0.584) acf

B1

2

(b) (i) $\frac{20}{120} \times \frac{19}{119} = \frac{19}{714} = 0.0266$
Method – allow with replacement
0.02655-0.02665 acf

M1 A1

2

(ii) $2 \times \frac{50}{120} \times \frac{20}{119} = \frac{50}{357} = 0.140$

2

Method – allow with replacement, 2 not required

= 0.1395 – 0.1405 acf

A1 B1 M1

3

(c) $6 \times \frac{50}{120} \times \frac{20}{119} \times \frac{35}{118} = \frac{125}{1003} = 0.125$

6 (allow 3)

Correct method –

allow with replacement; disallow 3

0.124-0.125 acf

B1 M1 A1

3

(d) $\frac{50}{120} \times \frac{49}{119} = \frac{35}{204} = 0.172$

Method – allow with replacement

Completely correct method

0.171-0.172 acf

M1 m1 A1

3

[13]

Q24.

(a) (i) $0.25 \times 0.05 = 0.125$

Method

0.0125 acf cao

M1 A1

2

(ii) $0.25 \times 0.05 + 0.3 \times 0.2 + 0.45 \times 0.03$

Method for Khan or Heinrich

Summation of probabilities for each

Completely correct method

M1 M1 m1

$= 0.032$

0.032 acf cao

A1

4

- (b) (i) 0.009
0.009 *cao*

B1

- (ii) $0.02 \times 0.2 = 0.004$
Method
0.004 *acf cao*

M1 A1

- (iii) $0.25 \times 0.03 = 0.0075$
Method
0.0075 *acf cao*

M1 A1

5

- (c) $0.25 \times 0.009 + 0.3 \times 0.004 + 0.45 \times 0.0075$
= 0.006825
Method their answers to (b)
Completely correct method
0.006825 (0.0068 – 0.00685) *acf*

M1 m1 A1

3

[14]